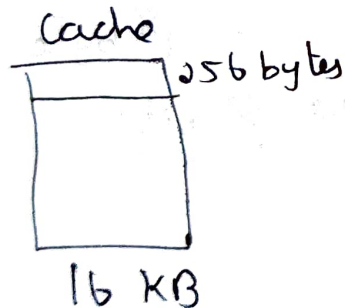


1. Consider a direct mapped cache of size 16 KB with block size ~~with block size~~ 256 bytes. The size of main m/y is 128 KB. Find -

1) No. of bits in tag

2) Tag directory size



Given,

cache memory size = 16 KB

Block size = Frame size = Line size = 256 bytes

Main mly size = 128 KB

we consider that the memory is byte addressable.

1) No: of bits in tag $\Rightarrow$ No: of bits in physical address
= Size of main memory.

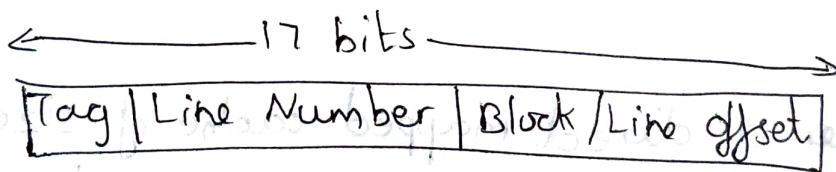
$$= 128 \text{ KB}$$

$$= 128 \times 1024$$

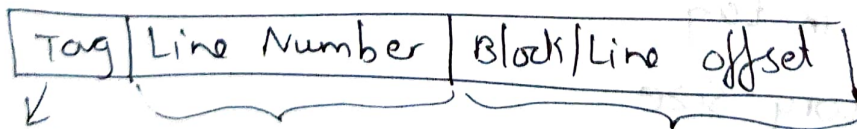
$$= 131072$$

$$= 2^{17}$$

$\therefore$ No: of bits in ^{physical address} ~~tag~~ = 17 bits



2)



17 - (6 + 8)
= 3 bits

Total no: of lines in
cache = $\frac{\text{Cache size}}{\text{Line size}}$

$$= \frac{16 \text{ KB}}{256 \text{ bytes}}$$

$$= \frac{16 \times 1024 \text{ bytes}}{256 \text{ bytes}}$$

$$= \frac{2^{14} \text{ bytes}}{2^8 \text{ bytes}} = 2^6 \text{ lines}$$

$\therefore$ No: of bits in line no: = 6 bits

Block size = 256

$$= 2^8 \text{ byte}$$

ie 8 bits

Tag Directory Size = No. of Tags $\times$ Tag size

or = No. of lines in cache $\times$ No. of bits in tag

$$\begin{aligned} & 2^6 \times 3 \\ &= 64 \times 3 \\ &= 192 \text{ bits} \\ &= \underline{\underline{24 \text{ bytes}}} \end{aligned}$$

$$= \frac{16 \text{ KB}}{256 \text{ bytes}} \times 3$$

$$= \frac{16 \times 1024 \text{ bytes}}{256 \text{ bytes}} \times 3$$

$$= 64 \times 3$$

$$= 192 \text{ bits}$$

$$= \underline{\underline{24 \text{ bytes}}}$$

$$8 \text{ bits} = 1 \text{ byte}$$

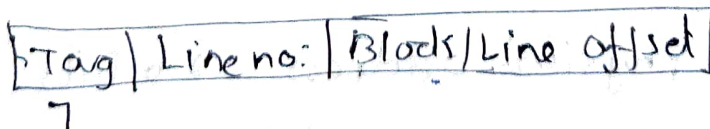
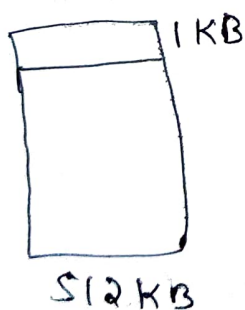
$$\therefore 192 \text{ bits} = \frac{192}{8}$$

$$= 24 \text{ bytes}$$

2. Consider a direct mapped cache of size 512 KB with block size 1 KB. There are 7 bits in the tag. Find

1. Size of the main memory

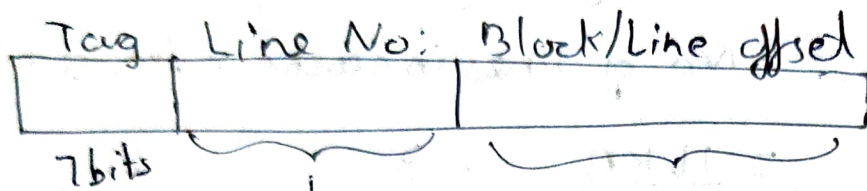
2. Tag directory size



Cache memory size = 512 KB

Block size = Frame size = Line size = 1 KB

No. of bits in tag = 7 bits



$$\frac{\text{cache Size}}{\text{Line Size}}$$

$$= \frac{512 \text{ KB}}{1 \text{ KB}}$$

$$= 512 \text{ lines}$$

$$= 2^9$$

$$\therefore \underline{9 \text{ bits}}$$

$$\text{Block size} = 1 \text{ KB}$$

$$= 1024 \text{ bytes}$$

$$= 2^{10} \text{ bytes}$$

$$\therefore \underline{10 \text{ bits}}$$

$$\begin{aligned} \text{Tag Directory Size} &= \text{No. of Tags} \times \text{Tag Size} \\ &= \text{No. of lines in cache} \times \text{No. of bits in tag} \end{aligned}$$

$$= 512 \times 7$$

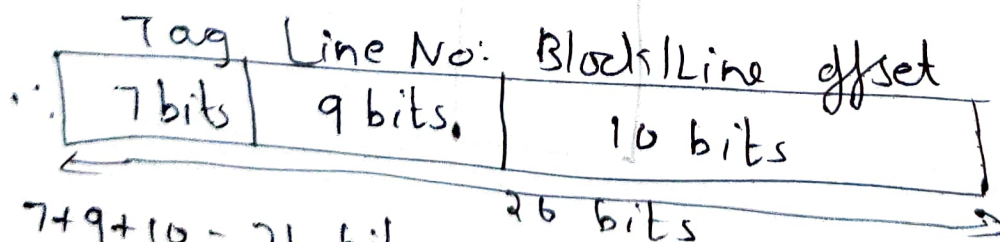
$$= 3584 \text{ bits}$$

$$= 448 \text{ bytes}$$

$$8 \text{ bit} = 1 \text{ byte}$$

$$3584 = 9$$

$$2^9 = 512$$



$$7 + 9 + 10 = 26 \text{ bits}$$

$$\text{No. of bits in physical address} = 26 \text{ bits}$$

$$\therefore \text{Size of main memory} = 2^{26} \text{ bytes}$$

$$\star \text{ Hit ratio} = \frac{\text{Hit}}{\text{Hit} + \text{Miss}} = \frac{\text{No. of hits}}{\text{Total access}}$$

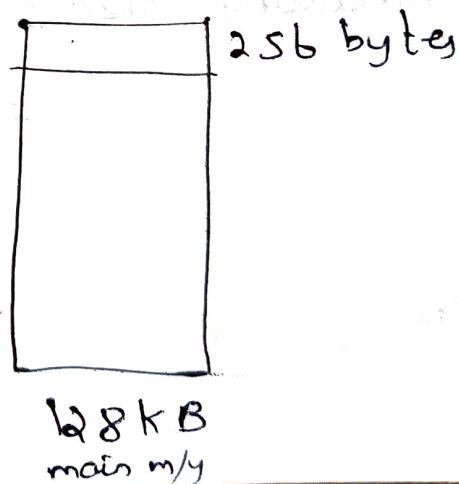
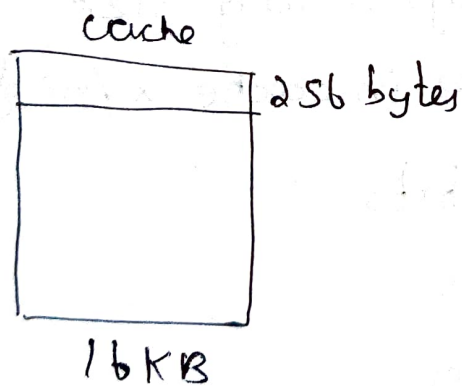
$$\text{Miss ratio} = \frac{\text{Miss}}{\text{Hit} + \text{Miss}} = \frac{\text{No. of miss}}{\text{Total access}} = 1 - \text{Hit ratio}$$

1. Consider a fully associative mapped cache of size 16 KB with block size 256 bytes. The size of main memory is 128 KB. Find

1. No. of bits in tag

2. Tag directory size

Ans



Cache memory size = 16 KB

Block size = Frame size = Line size = 256 bytes

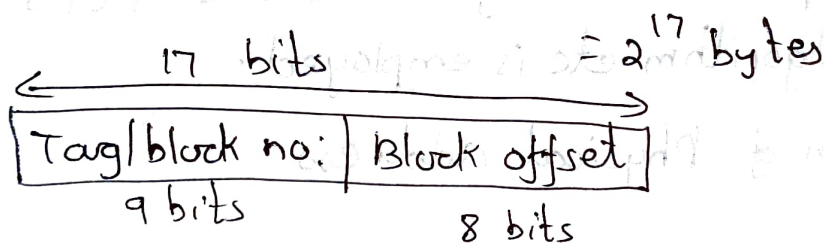
Main mly size = 128 KB

We consider that the main memory is byte addressable.

No. of bits in physical mly. = size of main mly
= 128 KB

= 128×1024 bytes

= 131072 bytes



No. of bits in block offset = Block size

= 256 bytes

= 2^8 bytes

ie 8 bits

No. of bits in Tag = $17 - 8 = \underline{\underline{9 \text{ bits}}}$

Tag Directory size = No. of tags $\times$ Tag size

= No. of lines in cache $\times$ No. of bits in Tag

= $2^6 \times 9 \text{ bits}$

= 576 bits

= 72 bytes

8 bit = 1 bytes

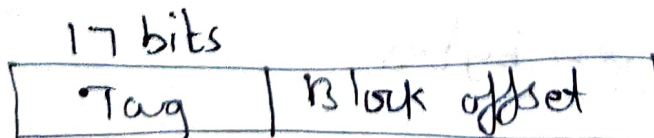
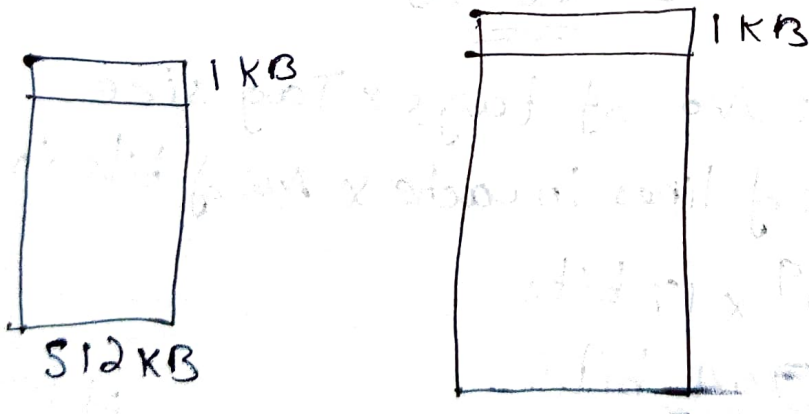
576 bits =

$$\begin{aligned}
 \text{No. of lines in cache} &= \frac{\text{Cache size}}{\text{Line size}} \\
 &= \frac{16 \text{ KB}}{256 \text{ bytes}} \\
 &= \frac{16 \times 1024 \text{ bytes}}{256 \text{ bytes}} \\
 &= \frac{16384}{256} \\
 &= 64 \text{ ~~bytes~~} = 2^6 \text{ lines}
 \end{aligned}$$

2. Consider a fully associative mapped cache of size 512 KB with block size 1 KB. There are 17 bits in the tag. Find:-

1. Size of main memory

2. Tag directory size.



Cache memory size = 512 KB

Block size = Frame size = ~~1~~ Line size = 1 KB

No. of bits in tag = 17 bits.

We consider that the memory is byte addressable.

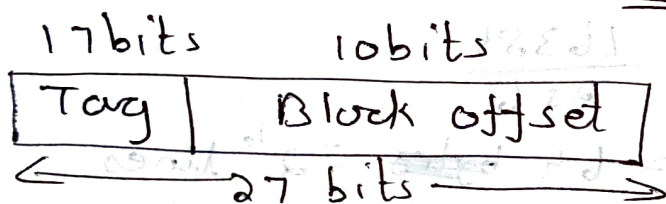
No. of bits in Block offset = Block size

$$= 1 \text{ KB}$$

$$= 1024 \text{ bytes}$$

$$= 2^{10} \text{ bytes}$$

ie 10 bits



Size of main memory = 2^{27} bytes

$$= \frac{134,217,728 \text{ bytes}}{1024}$$

$$= \frac{131,072 \text{ KB}}{1024}$$

$$= \underline{\underline{128 \text{ MB}}}$$

Tag directory size = No. of tags $\times$ Tag size

$$= \text{No. of lines in cache} \times \text{No. of bits in tag}$$

$$= 2^9 \times 17 \text{ bits}$$

$$= \frac{8704 \text{ bits}}{8}$$

$$= \underline{\underline{1088 \text{ bytes}}}$$

$$\begin{aligned} 8 \text{ bit} &= 1 \text{ byte} \\ 8704 \text{ bit} &= \end{aligned}$$

No: of lines in cache = $\frac{512 \text{ KB}}{1 \text{ KB}}$

$$= 512 \text{ lines} = \underline{\underline{2^9 \text{ lines}}}$$

Problems on Set Associative Cache

1. Consider a 2-way set associative mapped cache of size 16 KB with block size 256 bytes. The size of main memory is 128 KB. Find -

1. No. of bits in tag

2. Tag directory size

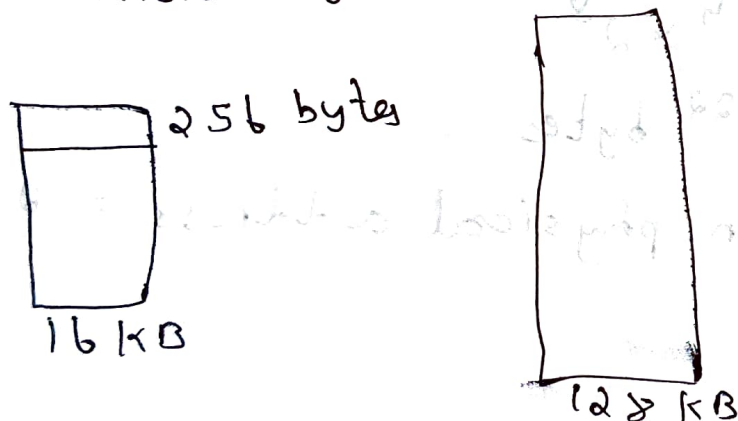
Ans. set size = 2

Cache memory size = 16 KB

Block size = Frame size = Line size = 256 bytes

Main mly size = 128 KB

We consider that the memory is byte addressable.



tag	block	word
Tag	Set Number	Block/Line offset

Size of main mly = 128 KB

= 128×1024 bytes

= $2^7 \times 2^{10}$ bytes

= 2^{17} bytes

$\therefore$ No. of bits in physical address = 17 bits

$$\begin{aligned}\text{No: of bits in block offset} &= \text{Block size} \\ &= 256 \text{ bytes} \\ &= 2^8 \text{ bytes}\end{aligned}$$

ie 8 bits

$$\text{Total no: of set in cache} = \frac{\text{Total no: of lines in cache}}{\text{set size}}$$

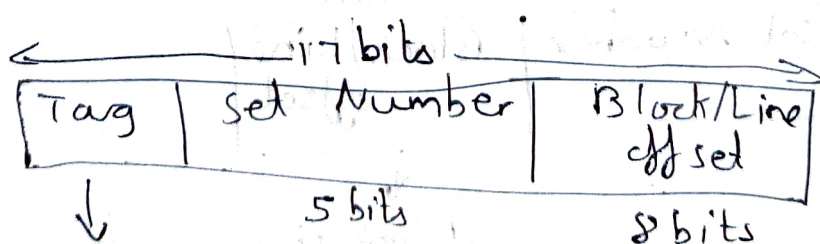
$$= \frac{\text{Cache Size}}{\text{Line Size}} \times \text{set size}$$

$$= \frac{16 \text{ KB}}{256 \text{ bytes}} \times 2$$

$$= \frac{16 \times 1024 \text{ bytes}}{256 \text{ bytes}} \times 2$$

$$= \frac{2^4 \times 2^{10} \text{ bytes}}{2^8 \text{ bytes}} \times 2 = \frac{2^6 \text{ lines}}{2} = \underline{\underline{2^5 \text{ sets}}}$$

$$\therefore \text{No: of bits in set number} = 5 \text{ bits}$$



$$\begin{aligned}\text{No: of bits in tag} &= 17 - (5 + 8) \\ &= 4 \text{ bits}\end{aligned}$$

$$\begin{aligned}\text{Tag Directory Size} &= \text{No: of tags} \times \text{Tag size} \\ &= \text{No: of lines in cache} \times \text{No: of bits in tag} \\ &= 2^6 \times 4 \text{ bits}\end{aligned}$$

$$= 64 \times 4 \text{ bits}$$

$$= 256 \text{ bits} = \frac{256}{8}$$

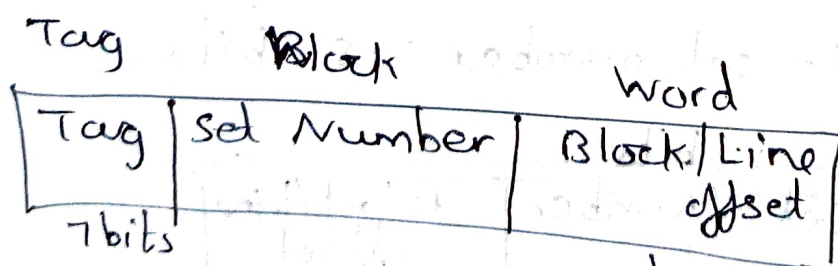
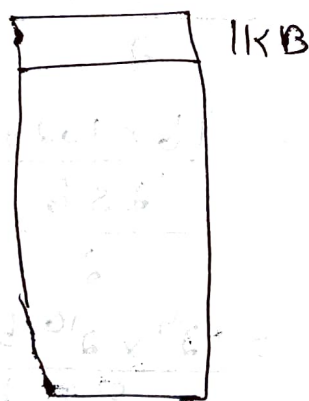
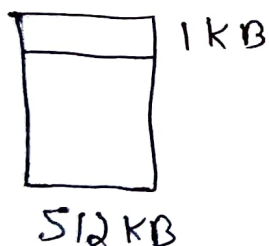
$$= \underline{32 \text{ bytes}}$$

$\therefore$ size of tag directory = 32 bytes

2. Consider a 8-way set associative mapped cache of size 512 KB with block size 1 KB. There are 7 bits in the tag. Find.

1. Size of main memory

2. Tag directory size



Block size = 1 KB

$$= 1 \times 1024 \text{ bytes}$$

$$= 2^{10} \text{ bytes}$$

$\therefore$ No. of bits in block offset = 10 bits

$$\text{Total no. of set in cache} = \frac{\text{Total no. of lines in cache}}{\text{Set size}}$$

$$= \frac{\text{Cache Size}}{\frac{\text{Line size}}{\text{Set size}}}$$

$$= \frac{512 \text{ KB}}{\frac{1 \text{ KB}}{8}}$$

$$= \frac{512 \times 1024 \text{ bytes}}{1 \times 1024 \text{ bytes}} \times \frac{2^9 \text{ lines}}{2^3} = 2^6 \text{ sets}$$

$$\therefore \text{Number of bits in set Number} = \underline{\underline{6 \text{ bits}}}$$

$$\begin{aligned} \text{No. of bits in physical address} &= 7 + 6 + 10 \\ &= 23 \text{ bits} \end{aligned}$$

$$\text{Size of main memory} = 2^{23} \text{ bytes}$$

$$= \frac{2^{23}}{1024 \times 1024} \text{ MB}$$

$$= \frac{2^{23}}{2^{10} \times 2^{10}} \text{ MB}$$

$$= 2^3 \text{ MB} = \underline{\underline{8 \text{ MB}}}$$

$$\text{Tag Directory Size} = \text{No. of Tags} \times \text{Tag Size}$$

$$= \text{No. of lines in cache} \times \text{No. of bits in tag}$$

$$= 2^9 \times 7 \text{ bits}$$

$$= 512 \times 7 \text{ bits}$$

$$= \underline{3584} \text{ bits}$$

$$= \underline{\underline{448}} \text{ bytes}$$